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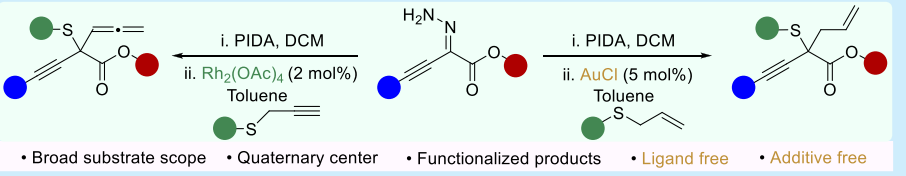
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Doyle–Kirmse Reaction on Alkynyl Hydrazone Carboxylates: Synthesis of 1,4-Allenyne and 1,5-Enyne Thioaryl Carboxylates

Himani Vaid, Akashdeep Sharma, Paru Jamwal, Pryanka Sharma, and Ramani Gurubrahamam*

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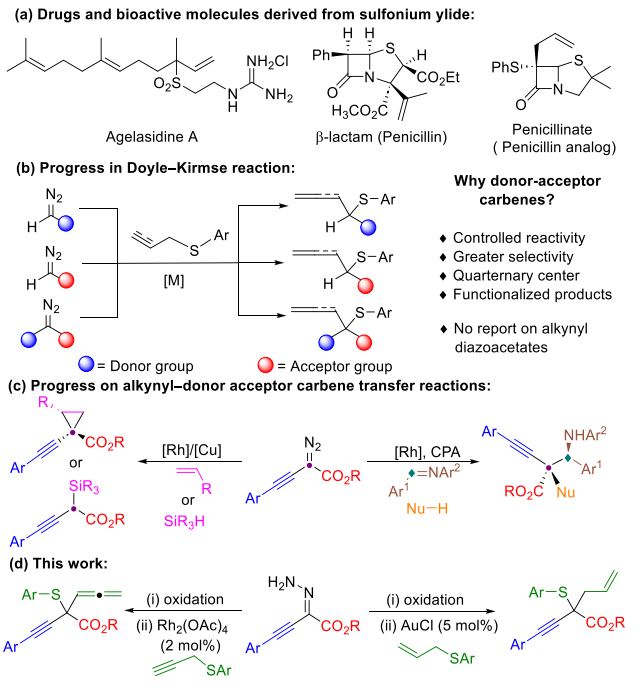
ABSTRACT: The first Doyle–Kirmse reaction on alkynyl diazoacetates using allyl/propargyl sulfides is reported. The development provides diversified 1,5-enyne and 1,4-allenyne thioaryl carboxylates in good yields under ligand-/additive-free AuCl and $Rh_2(OAc)_4$ catalysis, respectively (48 examples, up to 96% yield). The study demonstrated the dual role of allyl sulfide as a ligand and substrate. Also, we have exemplified various synthetic modifications of the products to showcase the utility of different functional groups.

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[2,3]-Sigmatropic rearrangement of sulfonium ylides has garnered significant interest among the synthetic community as it constructs C–C and C–S bonds simultaneously and provides various biologically active molecules (Scheme 1a).[1] The rearrangement of sulfonium ylides generated from diazo compounds and allyl/propargyl sulfides is popularly known as the Doyle–Kirmse reaction.[2] This reaction has been studied on different types of donor,[3] acceptor,[4] and donor–acceptor[5] diazo compounds, and a generalization of the reports is depicted in Scheme 1b.[6] The donor–acceptor carbenes are of paramount significance due to their controlled reactivity, high selectivity, and their ascendancy over the other carbenes to provide highly functionalized motifs with quaternary centers.[7,8] It is witnessed that the target-oriented synthesis of bioactive molecules containing quaternary centers is a formidable challenge in organic synthesis. Remarkably, the sigmatropic rearrangements contribute to about half the volume of such C–C bond-forming reactions by overcoming the steric challenges.[8]

From this perspective, the reports on sigmatropic rearrangement of carbene precursors having versatile π-bonds are relatively scarce. We noticed that [2,3]-sigmatropic rearrangement on alkynyl diazoacetates still remains scarce in the chemistry of donor–acceptor carbenes. In fact, the reactivity of alkynyl carbenes is still in its infancy, although they have been known for the past six decades.[9a] Probably, this could be due to their tendency to undergo rearrangement and isomerization.[9b] Recently, alkynyl carbene transfer reactions from donor or donor–donor type diazo compounds have been successfully witnessed in cyclopropanations[10] as well as Si–H,[11] B–H,[12] and C–C[13] bond insertion reactions. Also, a sigmatropic rearrangement reaction emerged through a decomposition of *N*-nosylhydrazones.[14] Noticeably, donor or

Scheme 1. Progress in Doyle–Kirmse Reaction



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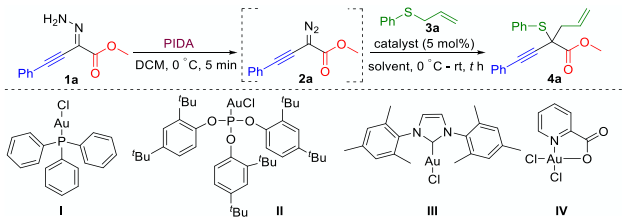
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donor−donor diazo analogues are unstable and are being generated *in situ* by a base-promoted decomposition of specially protected alkynyl hydrazones.[10−14] On the other hand, alkynyl carbene transfer reactions of alkynyl diazoacetates have merely been limited to cyclopropanations,[15,16] silane insertions,[16] and a multicomponent reaction[17] (Scheme 1c). In continuation of our research interest in alkynyl carbene transfer reactions,[16] herein, we are reporting the Doyle−Kirmse reaction on alkynyl donor−acceptor carbenes by employing allyl/propargyl aryl sulfides using transition metal catalysis (Scheme 1d). The development provides expedient access to highly functionalized and diversified 1,5-enyne and 1,4-allenyne thioaryl carboxylates.

We started our optimization studies by using alkynyl diazoacetate **2a** and allyl phenyl sulfide **3a** under coinage metal catalysis at 0 °C (Table 1, entries 1−7).[18] Encouraged

Table 1. Optimization Studies[a]



entry	1a/2a	Cat.	solvent	*t*/h	yield (%)[b]
1	**2a**	$Cu(MeCN)_4PF_6$	DCM	4	41
2	**2a**	$AgNO_3$	DCM	4	25
3	**2a**	AuCl	DCM	4	76
4	**2a**	I	DCM	4	70
5	**2a**	II	DCM	4	30
6	**2a**	III	DCM	4	16
7	**2a**	IV	DCM	4	67
8	**2a**	$Rh_2(OAc)_4$	DCM	4	60
9	**2a**	AuCl	$CHCl_3$	4	76
10	**2a**	AuCl	MeCN	4	74
11	**2a**	AuCl	toluene	4	78
12	**2a**	AuCl	toluene	6	81[c]
13[d]	**2a**	AuCl	toluene	6	74[c]
14[e]	**1a**	AuCl	toluene	6	56[c]
15[f]	**1a**	AuCl	toluene	6	73[c]
16[g]	**1a**	AuCl	toluene	6	88[c]

[a]Alkynyl diazoacetate **2a** (0.2 mmol, 1.0 equiv) was added dropwise to allyl phenyl sulfide **3a** (0.2 mmol, 1.0 equiv) and catalyst (5 mol %) in the given solvent at 0 °C. [b]Determined by crude ^{1}H NMR analysis using $CHPh_3$ as internal standard. [c]Isolated yield. [d]**2a** was added at rt. [e]1.0 equiv of **1a**. [f]1.2 equiv of **1a**. [g]1.5 equiv of **1a**. PIDA = Phenyliodine(III) diacetate.

by the success of gold catalysis in carbene transfer reactions,[19] we screened AuCl and its complexes of phosphine, phosphite, and NHC ligands. Delightfully, an additive- and ligand-free AuCl catalysis afforded the desired product **4a** in superior yield (76%, entry 3). Next, we screened the prominent $Rh_2(OAc)_4$ catalyst, and the reaction afforded the product but with a lower yield (60%, entry 8). Motivated by the superior yield with ligand- and additive-free AuCl catalysis, we further optimized the yield by screening different solvents and observed an improvement in toluene (81% yield, entries 9−12). Next, we screened temperature to improve the yield, but a reaction at ambient temperature did not improve the yield (entry 13). Finally, we attempted a one-pot synthesis from alkynyl hydrazone **1a** that could bypass the purification of diazoacetates **2a** after a metal-free oxidation with PIDA.[16] A reaction with hydrazone **1a** afforded the desired product **4a**; however, we observed a diminished yield (56%, entry 14). Delightfully, an improvement in yield (up to 88%) was noticed with an increased loading (1.0 to 1.5 equiv) of hydrazone **1a** (entries 15−16).

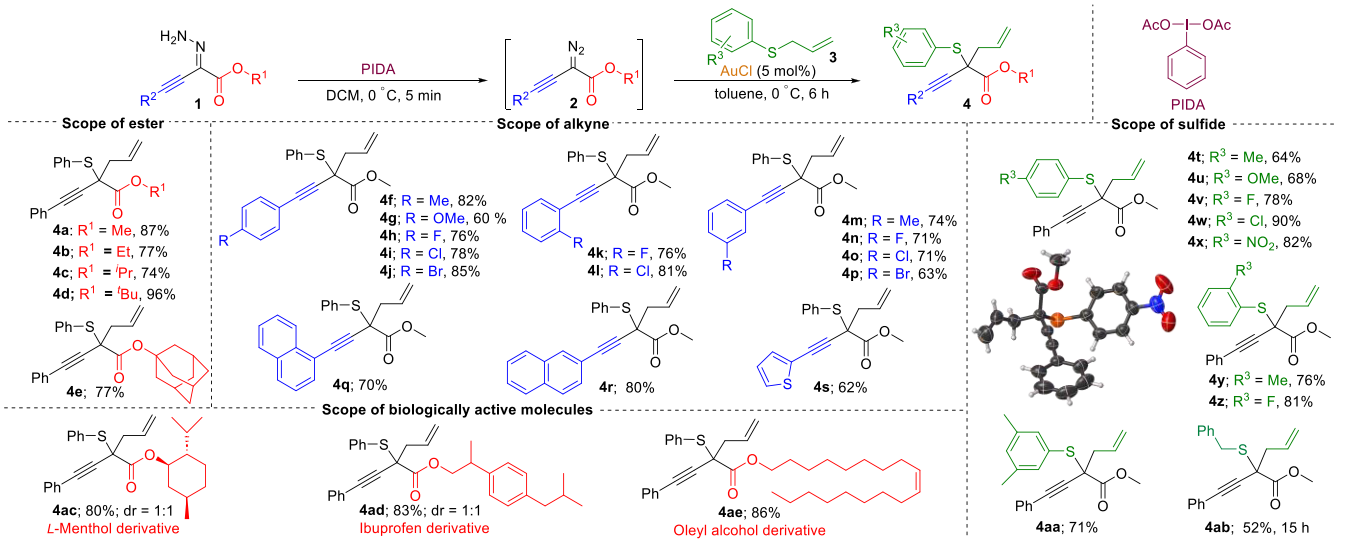
Having the optimized conditions in hand, we executed the substrate scope by varying the hydrazones **1** and sulfides **3** to obtain a wide variety of 1,5-enyne thioaryl carboxylates (Scheme 2). First, we examined the compatibility of the optimized conditions with the steric factor (methyl to adamantyl) of the ester group of alkynyl hydrazones. Delightfully, the reactions afforded the desired products **4a**−**4e** in good to excellent yields (74−96%). Then, we executed the alkyne scope by screening various hydrazones derived from *p*-, *o*-, and *m*-substituted aryl acetylenes. Gratifyingly, the reactions furnished the corresponding products **4f**−**4p** in moderate to good yields (60−85%). Further, we extended the scope by validating the hydrazones derived from naphthyl and thienyl acetylenes. The reactions furnished the corresponding products **4q**−**4s** in 62−80% yield. Next, we focused our attention toward the screening of electronic parameters on the aryl component of sulfide **3**. Electron-donating groups (Me and OMe) at the *p*-position afforded **4t**−**4u** in moderate yields (64−68%), while withdrawing groups (F, Cl, and NO_2) delivered the corresponding products **4v**−**4x** in good to excellent yields (78−90%). We screened *o*-substituted and polysubstituted thiols and obtained products **4y**−**4aa** in good yields (71−81%). We also screened a nonaryl substituted allyl sulfide and obtained the desired product **4ab** in 52% yield, however, after a longer reaction time. Finally, we executed the reactions with hydrazones derived from naturally available or biologically active molecules and observed the products **4ac**−**4ad** in good yields (80−83%), albeit with poor diastereoselectivity (1:1 dr). Further, we screened a hydrazone derived from oleyl alcohol and observed the product **4ae** in 86% yield.

Next, we extended our strategy to obtain 1,4-allenyne thioaryl carboxylates **6** by switching to propargyl sulfides **5** (Scheme 3). We carried out a brief optimization study and found that the reactions with the $Rh_2(OAc)_4$ catalyst outperformed that of AuCl.[18] The resultant highly functionalized 1,4-allenyne thioaryl carboxylate **6a** was obtained in 81% yield under the optimized conditions. We executed a short substrate scope by validating different hydrazones and aryl propargyl sulfides. As anticipated, the optimized conditions were compatible with the steric bulkiness of the ester and different aryl alkynes of the hydrazones. The resultant products **6b**−**6j** were obtained in good yields. Further, we extended the scope of an alkyne containing a fragile TIPS group. Delightfully, the desired product **6k** was obtained in 75% yield. Finally, we varied the aryl group of propargyl sulfides and obtained the products **6l**−**6p** in good yields (60−86%).

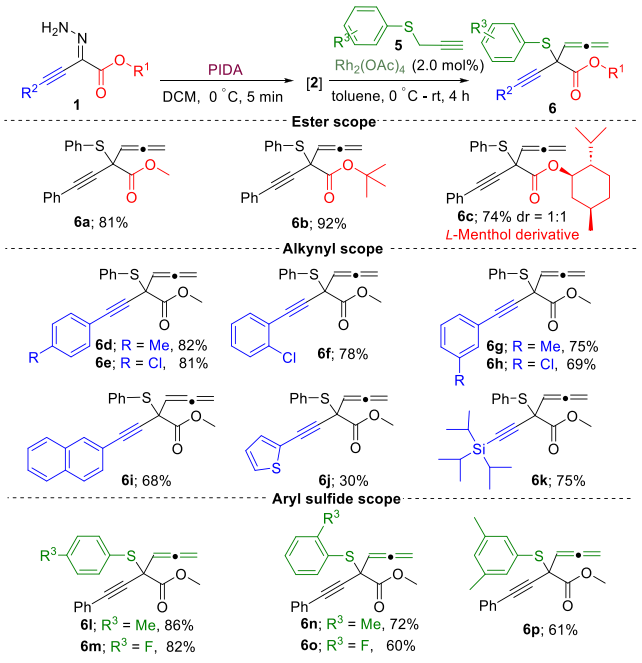
Next, we attempted a few control experiments to gain more insight into the reaction mechanism. First, we carried out reactions with benzyl and geranyl phenyl sulfides to differentiate the [2,3]-rearrangement from a probable [1,2]-rearrangement (Scheme 4a-4b). A reaction of **2a** with benzyl phenyl sulfide **7** did not provide the product **8** under the optimized conditions that could result *via* a [1,2]-sigmatropic rearrangement. In contrast, a reaction with geranyl sulfide **3k** led to **4af**, where a shift to the C3-carbon was noticed. This

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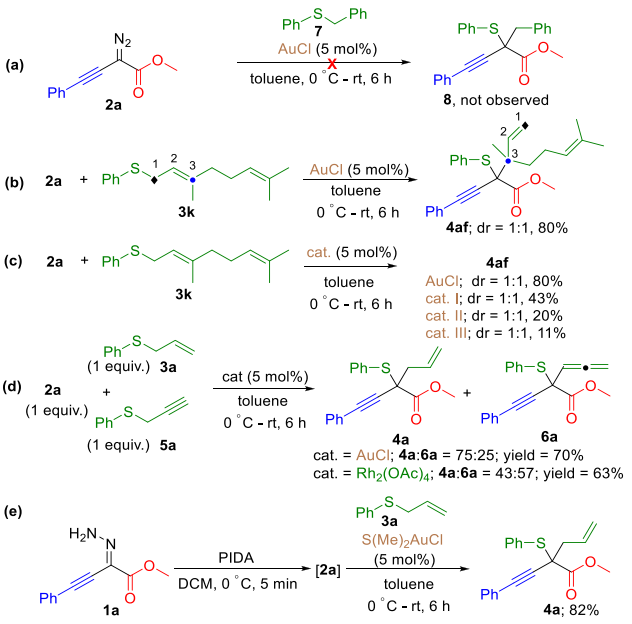
Scheme 2. Substrate Scope for 1,5-Enyne Thioaryl Carboxylates



Scheme 3. Substrate Scope for 1,4-Allenyne Thioaryl Carboxylates



Scheme 4. Control Experiments



clearly shows that the [2,3]-rearrangement is involved. Next, we intend to execute a few control experiments to probe whether the rearrangement is occurring through a metal-bound or metal-free sulfur ylide.[20] To investigate it, we executed a few reactions with **2a** and geranyl sulfide **3k** with different gold catalysts (Scheme 4c). The diastereoselectivity of the product in all of the reactions is found to be 1:1, irrespective of the gold catalyst and its associated ligands. It concludes that the rearrangement could occur through a metal-free ylide intermediate.[5d] Further, to understand the difference in reactivity of gold and rhodium carbenes with allyl and propargyl sulfides, we attempted a competitive experiment by employing **2a** and a mixture of **3a** and **5a** in the presence of gold/rhodium catalysis (Scheme 4d). We observed that gold carbenes are more reactive and have better compatibility with allyl sulfide **3a**, as the reaction with AuCl resulted in **4a**:**6a** in 70% yield with a 75:25 ratio. This observation is consistent with the literature precedent where the gold carbenes showed better reactivity with allyl sulfides than that of propargyl sulfides.[5d] While the rhodium carbenes are comparatively less reactive and have better compatibility with propargyl sulfide, the reaction with $Rh_2(OAc)_4$ provided **4a**/**6a** in a 43:57 ratio and 63% yield. During our studies, we observed a yellow coloration upon addition of sulfides to a colorless solution of AuCl. This could be attributed to a plausible coordination of sulfide with AuCl. Indeed, it was studied by ^{1}H NMR and HRMS analysis.[18] Delightfully, a reaction with the AuCl-(SMe_2) catalyst has afforded the product **4a** in a comparable yield (82%, Scheme 4e).[21] It is inferred that allyl sulfide plays a vital role in the reactions and acts as both the substrate and the ligand.

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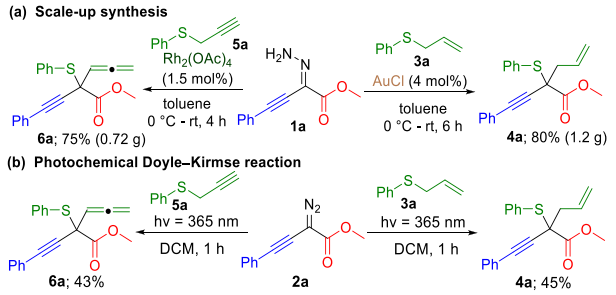
We also validated the optimized reaction conditions by carrying out scale-up reactions for the synthesis of 1,5-enyne and 1,4-allenyne thioaryl carboxylates (Scheme 5a). Gratifyingly, the conditions are compatible, and the desired products **4a** and **6a** were obtained in 80% and 75% yield, respectively. Encouraged by the recent research progress in visible-light-promoted metal-free Doyle−Kirmse reactions,[22] we tried to synthesize 1,5-enyne and 1,4-allenyne thioaryl carboxylates under photochemical conditions (Scheme 5b). After a brief optimization, we observed the products **4a** and **6a**, however, in moderate yields.[18]

Further, we attempted a few synthetic group transformations on densely functionalized products **4a**/**6k** (Scheme 6). First, we carried out functional group interconversions on the carboxylate group of **4a**. The carboxylic acid **9** and alcohol **10** derivatives were obtained after hydrolysis and reduction steps, respectively. Next, we attempted a hydration of alkynes that could generate 1,4-dicarbonyl compounds, which are excellent synthons for numerous transformations. To our delight, we obtained a highly valuable and functionalized 1,4-dicarbonyl core **11** in 60% yield upon treating **4a** with tosic acid while the quaternary center remained intact. Surprisingly, the reactions of **4a** with PIFA and I_2 have afforded the respective tri- and tetrasubstituted alkene cores (**12**−**13**) with excellent stereoselectivity.[18] Next, a desilylation reaction was executed on **6k** that resulted in the desired product **14** with a versatile terminal alkyne group.

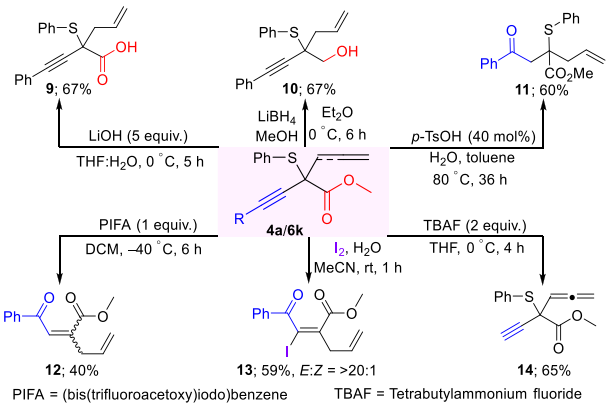
A tentative mechanism for the transition-metal-catalyzed Doyle−Kirmse reaction is drawn based on the literature and our observations (Scheme 7).[5d,20] First, unprotected alkynyl hydrazone **1** is oxidized to alkynyl diazoacetate **2** under PIDA promotion. An in situ generated gold-sulfide complex **15** decomposes the diazo compound **2** and forms reactive gold carbene **16**. Next, nucleophilic sulfide **3** adds to the electrophilic carbene and forms gold-bound ylide **17**. Subsequently, the gold-bound ylide will be stabilized by the sulfide **3** as in **TS1** which could dissociate into free ylide **18** and a gold-sulfide complex.[20c] Finally, ylide **18** undergoes a facile [2,3]-sigmatropic rearrangement through **TS2** that gives functionalized 1,5-enyne thioaryl carboxylates **4**.

In summary, we have shown the first [2,3]-sigmatropic rearrangement on alkynyl diazoacetate using gold/rhodium catalysis. The bench-stable alkynyl hydrazones were effectively employed and delivered 1,5-enyne/1,4-allenyne thioaryl carboxylates in good to excellent yields. The established protocol demonstrated a wide substrate scope and good functional group tolerance. Further, scale-up reactions and metal-free photochemical synthesis of propargyl sulfide carboxylates were demonstrated. In addition, the synthetic utility of the products was demonstrated, which delivered a functionalized 1,4-dicarbonyl alkene core and other motifs of interest. Lastly, we evaluated the role of the sulfide in a simple AuCl catalyzed Doyle−Kirmse reaction and found that sulfide acts as both ligand and substrate.

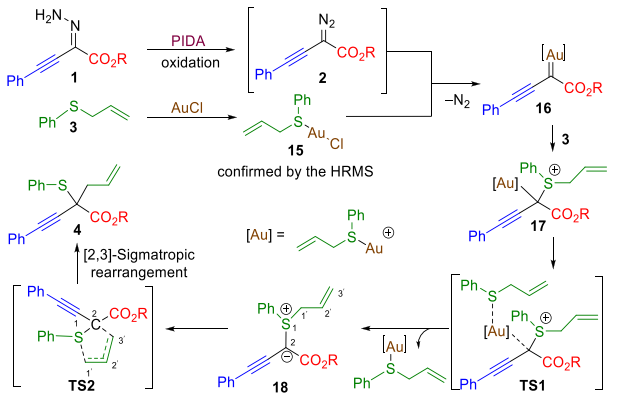
Scheme 5. Scale-up Synthesis and Photochemical Doyle−Kirmse Reaction



Scheme 6. Synthetic Transformations



Scheme 7. Plausible Mechanism



■ ASSOCIATED CONTENT

Data Availability Statement

The data underlying this study are available in the published article and its Supporting Information.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.4c00545.

General information; Materials; General procedures and characterization data; Control experiments; NMR spectra; Crystallographic information for **4x** (PDF)

FAIR data, including the primary NMR FID files, for compounds **4a**−**4af**, **6a**−**6p**, and **9**−**14** (ZIP)

Accession Codes

CCDC 2291743 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, or by emailing data_request@ccdc.cam.ac.uk, or by contacting The Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, UK; fax: +44 1223 336033.

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AUTHOR INFORMATION

Corresponding Author

Ramani Gurubrahamam − *Department of Chemistry, Indian Institute of Technology Jammu, Jammu and Kashmir 181221, India*; orcid.org/0000-0003-3035-1189; Email: guru.ramani@iitjammu.ac.in

Authors

Himani Vaid − *Department of Chemistry, Indian Institute of Technology Jammu, Jammu and Kashmir 181221, India*

Akashdeep Sharma − *Department of Chemistry, Indian Institute of Technology Jammu, Jammu and Kashmir 181221, India*

Paru Jamwal − *Department of Chemistry, Indian Institute of Technology Jammu, Jammu and Kashmir 181221, India*

Pryanka Sharma − *Department of Chemistry, Indian Institute of Technology Jammu, Jammu and Kashmir 181221, India*

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Notes

The authors declare no competing financial interest.

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